# Estructura y tablas en OMOP CDM

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**Real-World Epidemiology (RWEpi) Group**

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1. Bases de datos, SQL y R
2. Estructura OMOP CDM
3. Tablas de CDM OMOP
   - Tablas de datos clínicos
   - Tablas de vocabulario

# I. Bases de datos, SQL y R

*Estructura y tablas en OMOP CDM*

# I. Bases de datos, SQL y R

¿Qué es una base de datos (BBDD)?

SISTEMA DISEÑADO PARA **ALMACENAR, GESTIONAR Y RECUPERAR** DATOS DE MANERA **EFICIENTE**.

LA ESTRUCTURA DE LA BBDD DEPENDE DE LA **NATURALEZA** Y **VOLUMEN** DE LOS DATOS.

Algunas características básicas de las BBDD:

- MODELO ESTRUCTURAL DE LA BBDD
- DBMS: SOFTWARE P/ADMINISTRAR LA BBDD
- DEFINICIÓN DE ESQUEMAS (TABLAS, RELACIONES, RESTRICCIONES)

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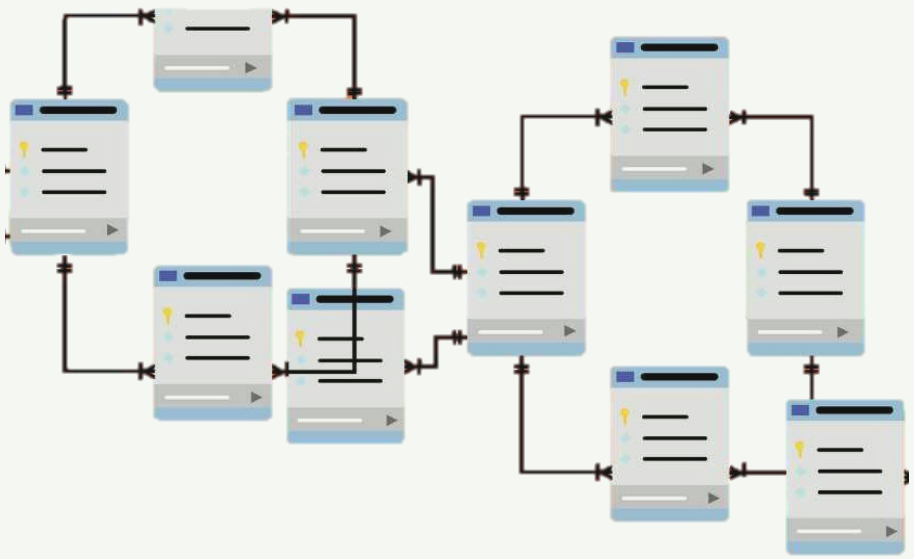
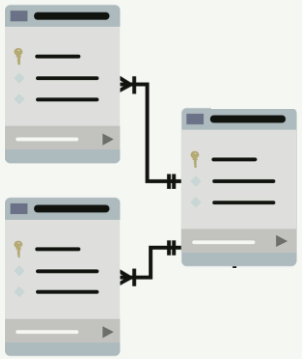
# I. Bases de datos, SQL y R

**Base de datos relacional**

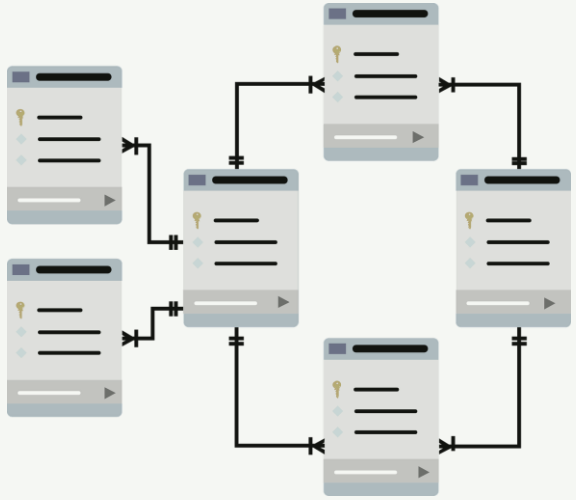
Esquema 1



Esquema 2



Esquema 3



# I. Bases de datos, SQL y R

**PROPÓSITO EN LA DEFINICIÓN DE ESQUEMAS:**

- ORGANIZAR LOS DATOS DE FORMA ESTRUCTURADA
- ACCESO SIMULTÁNEO SIN INTERFERIR EN LOS PROCESOS AJENOS
- INTEGRIDAD DE LA INFORMACIÓN

# I. Bases de datos, SQL y R

**SQL:** Lenguaje estándar para interactuar con la base de datos

Tipos de comandos en SQL:

1. DEFINICIÓN DE TABLAS (create, alter, drop tables)
2. MANIPULACIÓN DE DATOS (insert into, update, delete)
3. CONSULTA DE DATOS (select)
4. CONTROL DE DATOS

# I. Bases de datos, SQL y R

**SQL**

- 👍 Lenguaje estándar utilizado por las bases de datos.
- 👍 SQL está optimizado para manejar y manipular grandes volúmenes de datos.
- ☹ Problema: no tiene soporte para realizar operaciones estadísticas.

**R**

- 👍 Lenguaje de programación ampliamente utilizado en estadística.
- 👍 R es fuerte en la manipulación de datos, el análisis estadístico y la visualización.
- ☹ Problema: podría no ser eficiente por grandes bases de datos.

# I. Bases de datos, SQL y R

Combinar SQL y R para aprovechar las ventajas de cada una de las herramientas.

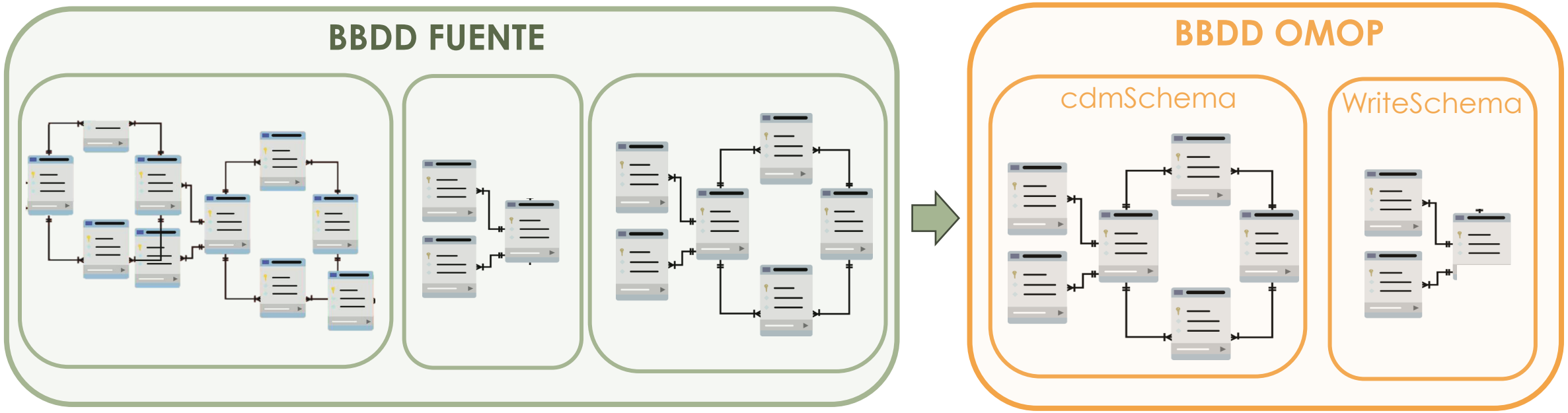
- **SQL:** creación y gestión de la base de datos, guardada en el **servidor**

- **R:**
  1. Conexión al servidor para acceder a la base de datos (BBDD).
  2. Manipulación y análisis de los datos (que estan en el servidor).
  3. Exportar del servidor a nuestro ordenador los resultados.

# 2. Estructura OMOP CDM

# 2. Estructura OMOP CDM

Modelo de datos común diseñado para **armonizar datos clínicos y administrativos** de diferentes fuentes de salud para optimizar los análisis observacionales en red.



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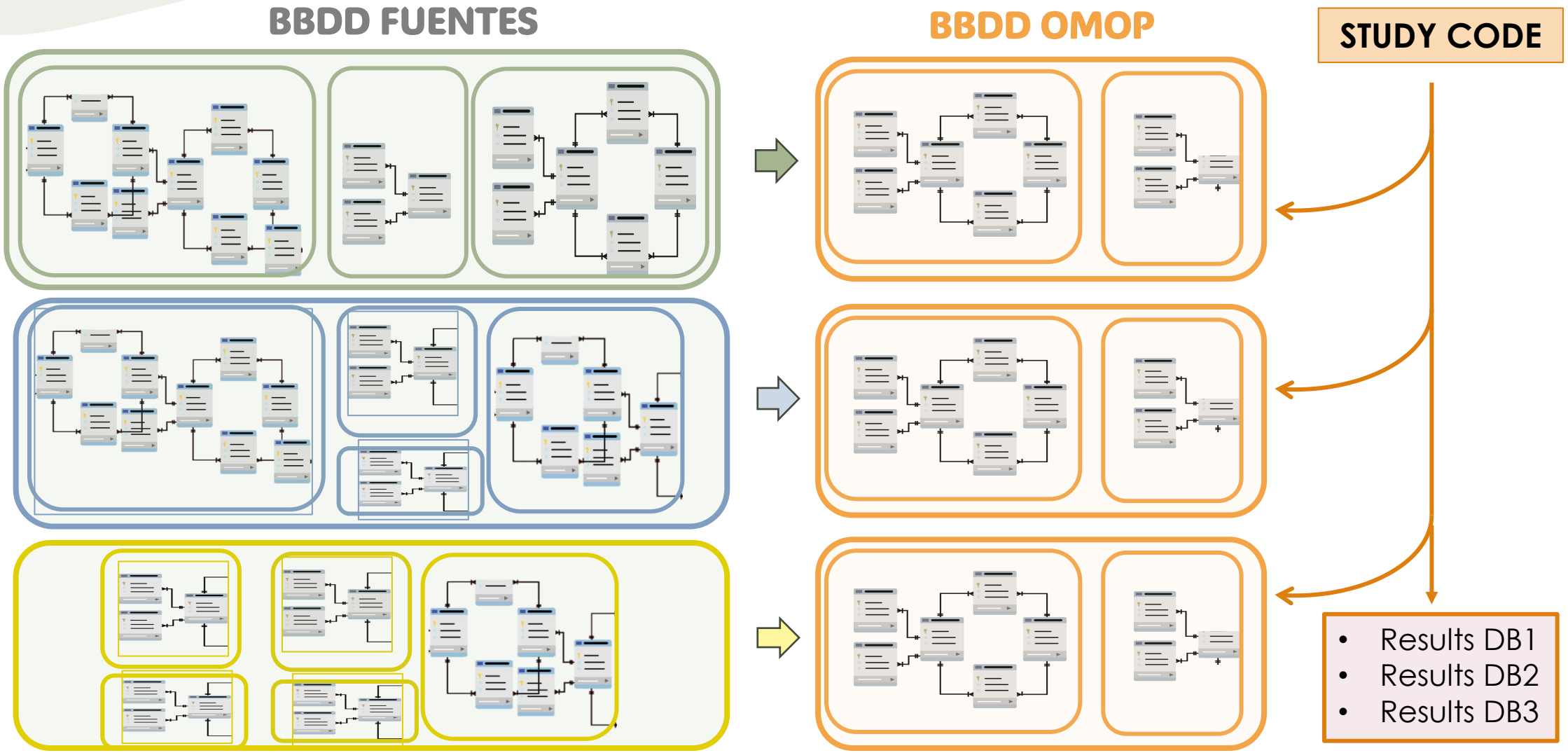
# 2. Estructura OMOP CDM



# 2. Estructura OMOP CDM

**Principios de diseño de OMOP**:

- Adecuación al propósito: optimizar análisis
- Protección de datos
- Diseño de dominios relacional (PERSON-center)
- Vocabularios estandarizados
- Re-utilización de vocabularios existentes
- Mantenimiento de códigos fuente
- Neutralidad tecnológica
- Escalabilidad
- Compatibilidad con versiones anteriores

# 2. Estructura OMOP CDM

**Convenciones en OMOP CDM**:

1. Hay **dos esquemas** en el CDM.
2. Los eventos de diferente naturaleza se **organizan por dominios.**
3. El contenido de las tablas se representa a través de **conceptos**.
4. El **nombre de los campos** sigue ciertas reglas.

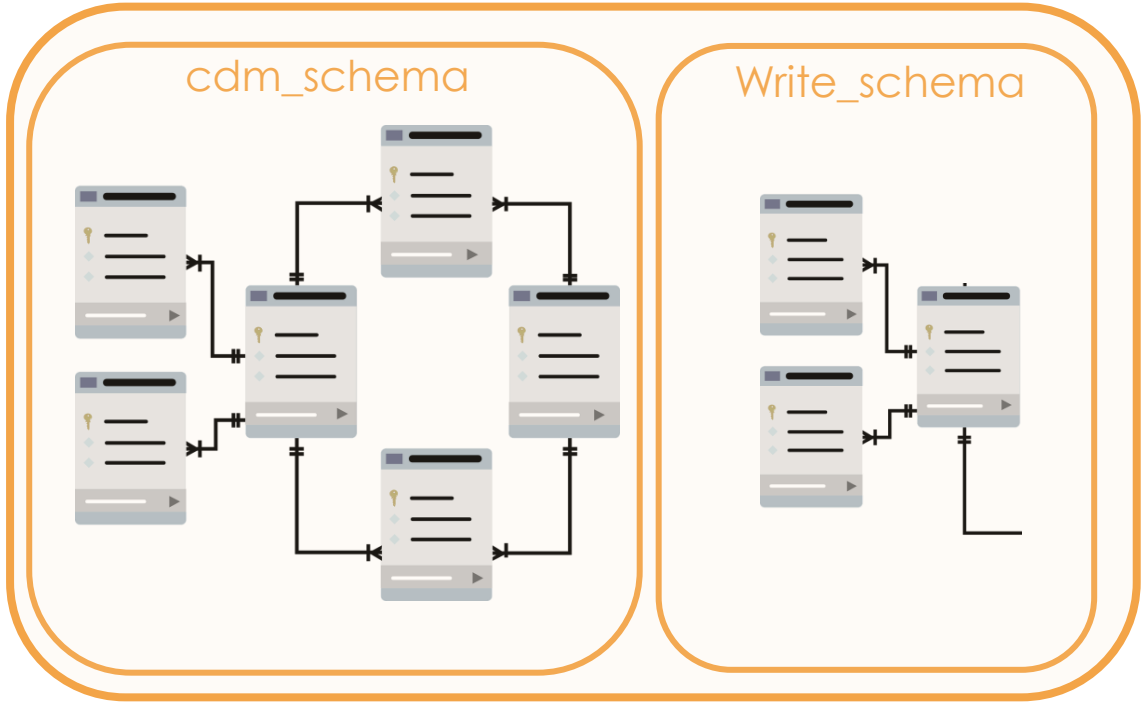
# 2. Estructura OMOP CDM

**Convenciones en OMOP CDM**:

1. Hay **dos esquemas** en el CDM:

cdmSchema: Sólo lectura. Contiene todas las tablas con datos clínicos, administrativos, vocabularios, etc.

writeSchema: Lectura/escritura. Podemos guardar tablas que hemos creado (por ej. cohortes).



# 2. Estructura OMOP CDM

**Convenciones en OMOP CDM**:

2. Los eventos de diferente naturaleza se **organizan por dominios:**

- Cada concepto estándar tiene un dominio único.
- El dominio define en qué tabla se registra el concepto.
- Correspondencia dominio-tabla-campo inequívoca.

# 2. Estructura OMOP CDM

**Convenciones en OMOP CDM**:

2. Los eventos de diferente naturaleza se **organizan por dominios:**

**UN EJEMPLO POR FAVOR!!!**

**acetaminophen (concept_id = 1125315 en RxNorm )**

Cada concepto estándar tiene un dominio único. → Concept **1125315**: Drug

El dominio define en qué tabla se registra el concepto. → Drug: Drug_exposure

Correspondencia dominio-tabla-campo inequívoca. → DRUG_CONCEPT_ID

# 2. Estructura OMOP CDM

**Convenciones en OMOP CDM**:

2. Los eventos de diferente naturaleza se **organizan por dominios:**

**UN EJEMPLO POR FAVOR!!!**

**acetaminophen (concept_id = 1125315 en RxNorm )**

| | drug_exposure_id | person_id | drug_concept_id | drug_exposure_start_date |
|---|---|---|---|---|
| 1 | 512096999 | 476544 | 1125315 | 2006-02-01 |
| 2 | 1163959837 | 476593 | 1125315 | 2009-03-01 |
| 3 | 738582891 | 476705 | 1125315 | 2010-06-01 |
| 4 | 727546402 | 476705 | 1125315 | 2009-04-01 |
| 5 | 698516767 | 476705 | 1125315 | 2008-12-01 |

# 2. Estructura OMOP CDM

**Convenciones en OMOP CDM**:

3. El contenido de las tablas se representa a través de **conceptos:**

**CONDITION_OCCURRENCE**

| | condition_occurrence_id | person_id | condition_concept_id | condition_start_date |
|---|---|---|---|---|
| 1 | 3831 | 4117 | 4220631 | 2020-04-09 |
| 2 | 8826 | 9595 | 317576 | 1990-07-18 |
| 3 | 2567 | 2866 | 317576 | 1952-12-01 |
| 4 | 5722 | 6246 | 381316 | 2012-05-25 |
| 5 | 1629 | 1905 | 434490 | 2020-12-31 |
| 6 | 9707 | 10526 | 317576 | 2010-01-01 |

# 2. Estructura OMOP CDM

## Convenciones en OMOP CDM:

3. El contenido de las tablas se representa a través de **conceptos:**

**CONDITION_OCCURRENCE**

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| 4 | 5722 | 6246 | 381316 | 2012-05-25 |
| 5 | 1629 | 1905 | 434490 | 2020-12-31 |
| 6 | 9707 | 10526 | 317576 | 2010-01-01 |

**CONCEPT**

| concept_id | concept_name | domain_id | vocabulary_id |
|---|---|---|---|
| 317576 | Coronary arteriosclerosis | Condition | SNOMED |

# 2. Estructura OMOP CDM

**Convenciones en OMOP CDM**:

4. El **nombre de los campos** sigue ciertas reglas:

| Notation | Description |
|---|---|
| [Event]_ID | Identificador único de cada registro |
| [Event]_CONCEPT_ID | Clave externa a un concepto estándar (CONCEPT) |
| [Event]_SOURCE_CONCEPT_ID | Clave externa a un concepto fuente (CONCEPT) |
| [Event]_TYPE_CONCEPT_ID | Clave externa sobre el origen del registro (CONCEPT) |
| [Event]_SOURCE_VALUE | Texto libre que refleja cómo se registró en la fuente |

# 3. Tablas en OMOP CDM

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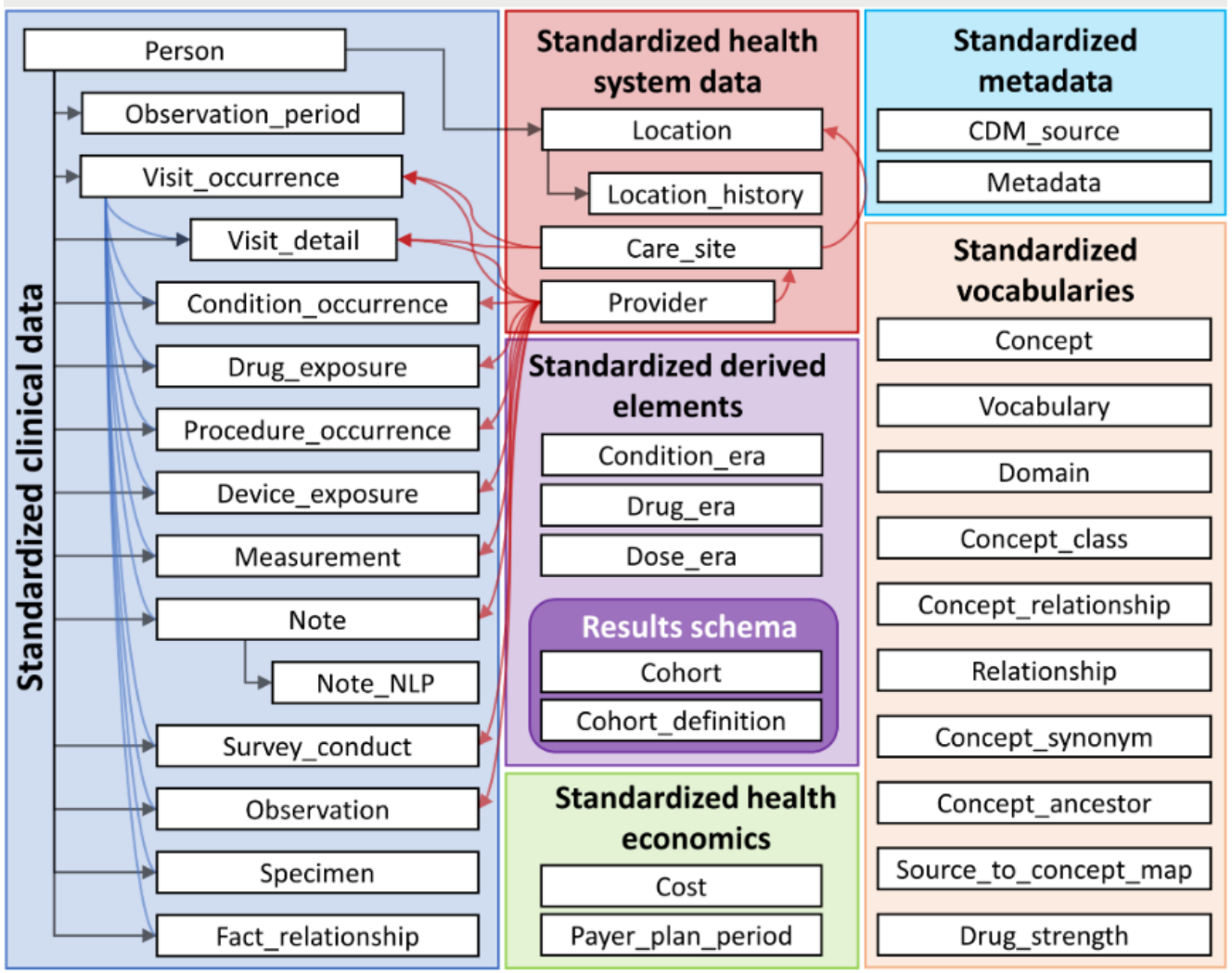
# 3. Tablas en OMOP CDM




THE BOOK OF OHDSI
OBSERVATIONAL HEALTH DATA SCIENCES AND INFORMATICS

https://ohdsi.github.io/TheBookOfOhdsi/

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# 3. Tablas en OMOP CDM

## DATOS CLÍNICOS

- PERSON🧑
- OBSERVATION_PERIOD 👀
- VISIT_OCCURRENCE🏥
- CONDITION_OCCURRENCE🤒
- DRUG_EXPOSURE💊
- DEVICE_EXPOSURE
- PROCEDURE_OCCURRENCE
- MEASUREMENT
- OBSERVATION

## METADATA

### DATOS DEL SISTEMA SANITARIO

LOCATION / CARE_SITE / PROVIDER

### DATOS DE VOCABULARIO

- CONCEPT
- CONCEPT_RELATIONSHIP
- CONCEPT_ ANCESTOR
- SOURCE_TO_CONCEPT_MAP

### DATOS COSTOS/RECURSOS

- COST
- PAYER_PLAN_PERIOD

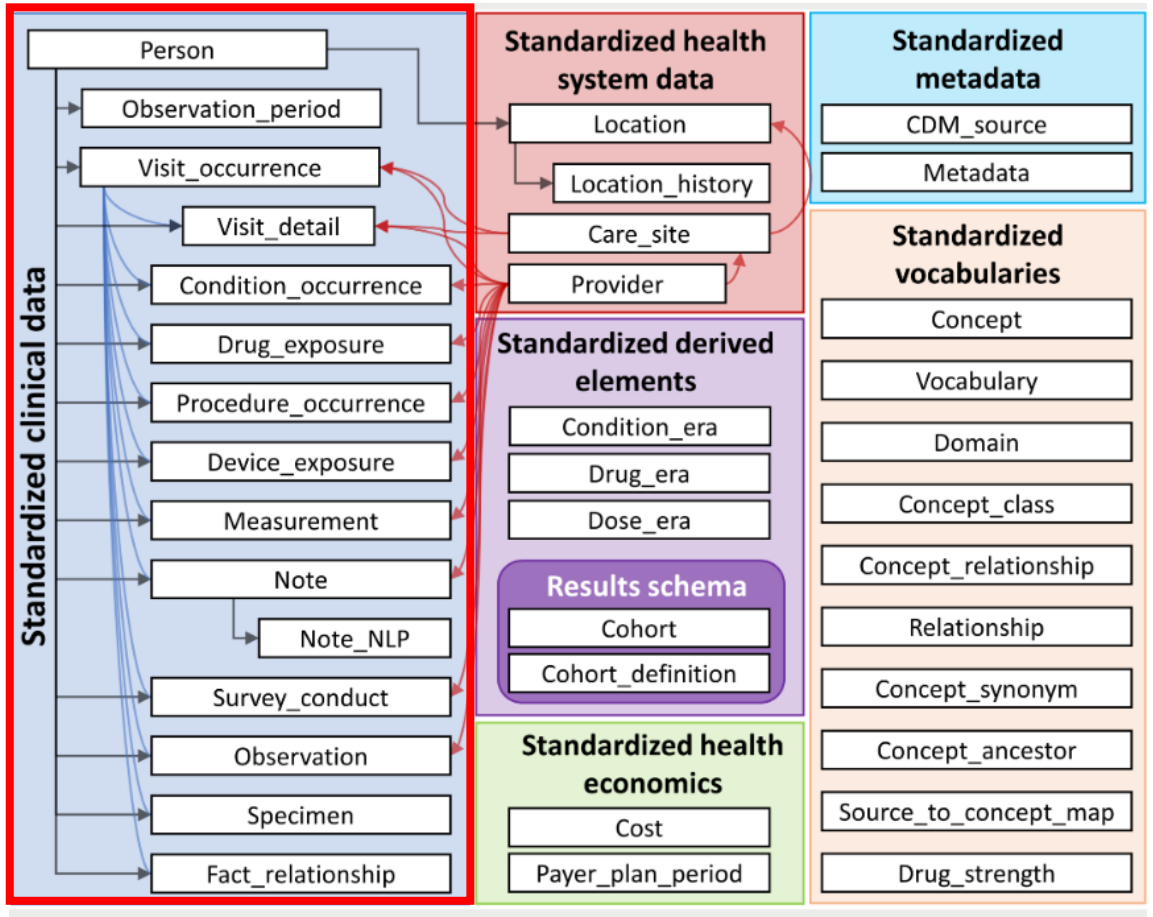
## DERIVADAS

- DRUG_ERA
- DOSE_ERA
- CONDITION_ERA

## WRITE SCHEMA

- **COHORTS**
- **TABLES**

# Tablas de datos clínicos en OMOP



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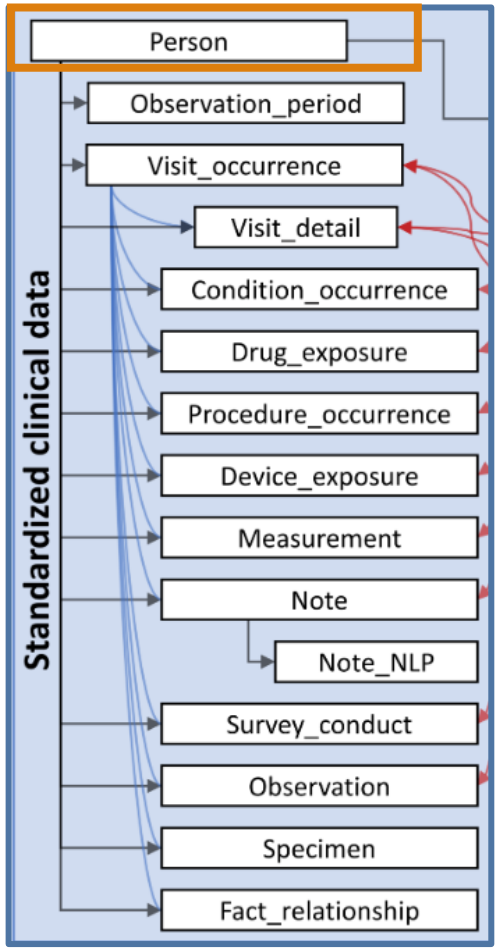
# Tablas de datos clínicos




**PERSON:**

```
$ person_id                  <int> ...
$ gender_concept_id          <int> ...
$ year_of_birth              <int> ...
$ month_of_birth             <int> ...
$ day_of_birth               <int> ...
$ birth_datetime             <dttm> ...
$ race_concept_id            <int> ...
$ ethnicity_concept_id       <int> ...
$ location_id                <int> ...
$ provider_id                <int> ...
$ care_site_id               <int> ...
$ person_source_value        <chr> ...
$ gender_source_value        <chr> ...
$ gender_source_concept_id   <int> ...
$ race_source_value          <chr> ...
$ race_source_concept_id     <int> ...
$ ethnicity_source_value     <chr> ...
$ ethnicity_source_concept_id <int> ...
```

Identificador único de la persona (clave primaria)

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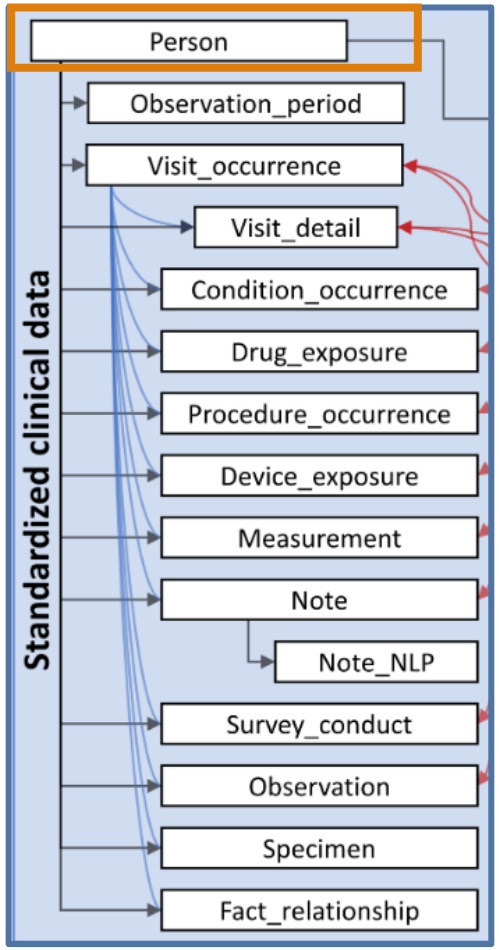
# Tablas de datos clínicos




```
> cdm$person
# Source:   table<main.person> [?? x 18]
# Database: DuckDB v0.10.2 [agiuliodori@Windows 10 x64:R 4.4.0/C:\Users\agiuliodori
942.duckdb]
```

| | person_id | gender_concept_id | year_of_birth | month_of_birth | day_of_birth | birth_date |
|---|---|---|---|---|---|---|
| | <int> | <int> | <int> | <int> | <int> | <dttm> |
| 1 | 1 | 8532 | 1970 | 4 | 24 | 1970-04-24 |
| 2 | 2 | 8532 | 1929 | 3 | 18 | 1929-03-18 |
| 3 | 3 | 8532 | 1970 | 4 | 4 | 1970-04-04 |
| 4 | 4 | 8507 | 1966 | 2 | 26 | 1966-02-26 |
| 5 | 5 | 8532 | 1936 | 6 | 10 | 1936-06-10 |
| 6 | 6 | 8507 | 1996 | 5 | 29 | 1996-05-29 |
| 7 | 7 | 8507 | 1923 | 11 | 14 | 1923-11-14 |
| 8 | 8 | 8507 | 2018 | 8 | 20 | 2018-08-20 |
| 9 | 9 | 8532 | 1933 | 2 | 11 | 1933-02-11 |
| 10 | 10 | 8507 | 2010 | 3 | 7 | 2010-03-07 |

Identificador único de la persona (clave primaria)

Clave externa a un concepto estándar

```
> cdm$concept |> filter(concept_id %in% c(8507, 8532))
# Source:   SQL [?? x 10]
# Database: DuckDB 1.4.4 [agiuliodori@Windows 10 x64:R 4.4.0/C
```

| | concept_id | concept_name | domain_id | vocabulary_id | concept_clas |
|---|---|---|---|---|---|
| | <int> | <chr> | <chr> | <chr> | <chr> |
| 1 | 8532 | FEMALE | Gender | Gender | Gender |
| 2 | 8507 | MALE | Gender | Gender | Gender |

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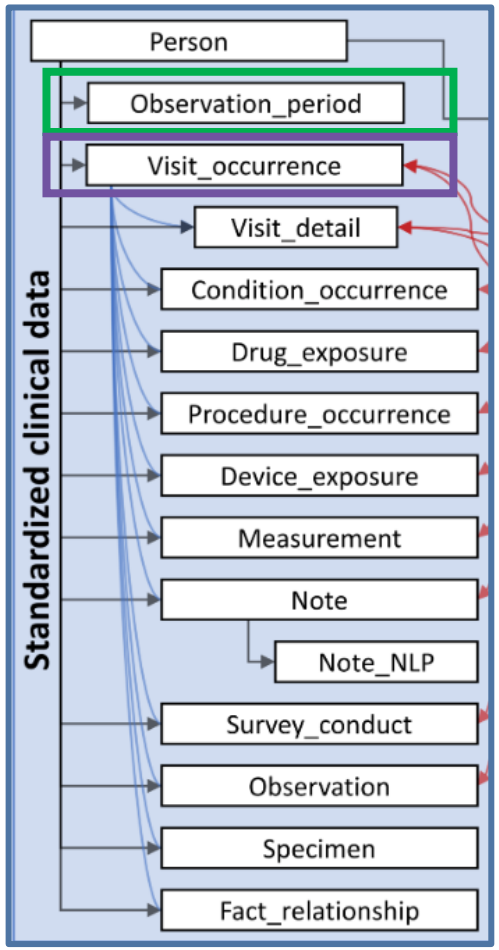
# Tablas de datos clínicos




## OBSERVATION_PERIOD:

```
$ observation_period_id          <int>
$ person_id                      <int>
$ observation_period_start_date  <date>
$ observation_period_end_date    <date>
$ period_type_concept_id         <int>
```

Id único del periodo de observación (clave primaria)
Clave externa a la persona

## VISIT_OCCURRENCE:

```
$ visit_occurrence_id            <int>
$ person_id                      <int>
$ visit_concept_id               <int>
$ visit_start_date               <date
$ visit_start_datetime           <dttm
$ visit_end_date                 <date
$ visit_end_datetime             <dttm
$ visit_type_concept_id          <int>
$ provider_id                    <int>
$ care_site_id                   <int>
$ visit_source_value             <chr>
$ visit_source_concept_id        <int>
$ admitting_source_concept_id    <int>
$ admitting_source_value         <chr>
$ discharge_to_concept_id        <int>
$ discharge_to_source_value      <chr>
$ preceding_visit_occurrence_id  <int>
```

Identificador único de la visita (clave primaria)
Clave externa a la persona

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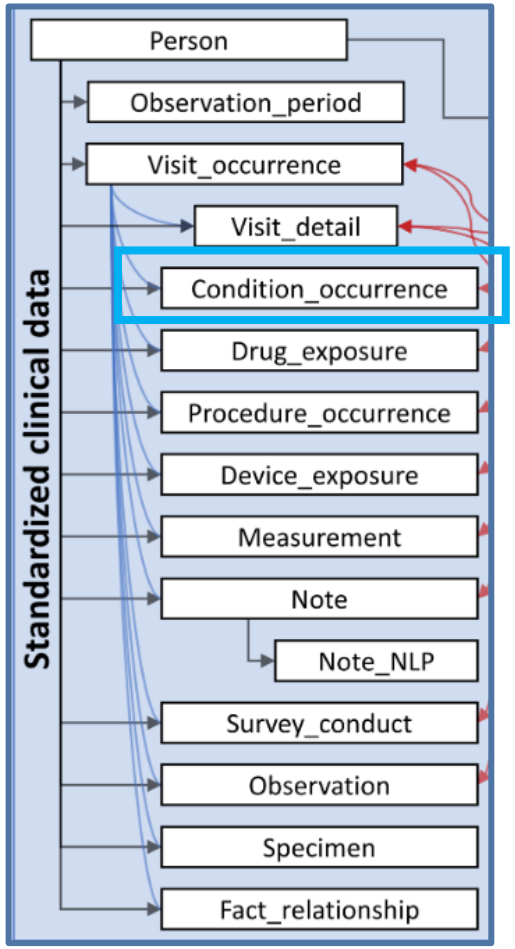
# Tablas de datos clínicos




## CONDITION_OCCURRENCE:

```
$ condition_occurrence_id      <int> .
$ person_id                    <int> .
$ condition_concept_id         <int> .
$ condition_start_date         <date>
$ condition_start_datetime     <dttm>
$ condition_end_date           <date>
$ condition_end_datetime       <dttm>
$ condition_type_concept_id    <int> .
$ condition_status_concept_id  <int> .
$ stop_reason                  <chr> .
$ provider_id                  <int> .
$ visit_occurrence_id          <int> .
$ visit_detail_id              <int> .
$ condition_source_value       <chr> .
$ condition_source_concept_id  <int> .
$ condition_status_source_value <chr> .
```

Clave primaria

Clave externa a la persona

Clave externa al concepto

Clave externa a la visita

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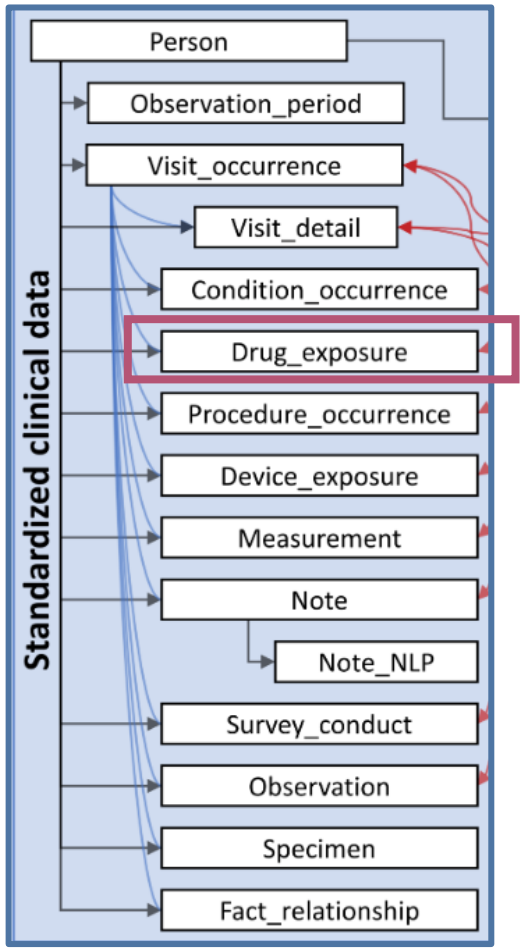
# Tablas de datos clínicos




## DRUG_EXPOSURE:

```
$ drug_exposure_id          <int> ..
$ person_id                 <int> ..
$ drug_concept_id           <int> ..
$ drug_exposure_start_date  <date>
$ drug_exposure_start_datetime <dttm>
$ drug_exposure_end_date    <date>
$ drug_exposure_end_datetime <dttm>
$ verbatim_end_date         <date>
$ drug_type_concept_id      <int> ..
$ stop_reason               <chr> ..
$ refills                   <int> ..
$ quantity                  <dbl> ..
$ days_supply               <int> ..
$ sig                       <chr> ..
$ route_concept_id          <int> ..
$ lot_number                <chr> ..
$ provider_id               <int> ..
$ visit_occurrence_id       <int> ..
$ visit_detail_id           <int> ..
$ drug_source_value         <chr> ..
$ drug_source_concept_id    <int> ..
$ route_source_value        <chr> ..
$ dose_unit_source_value    <chr> ..
```

Clave primaria

Clave externa a la persona

Clave externa al concepto

Clave externa a la visita

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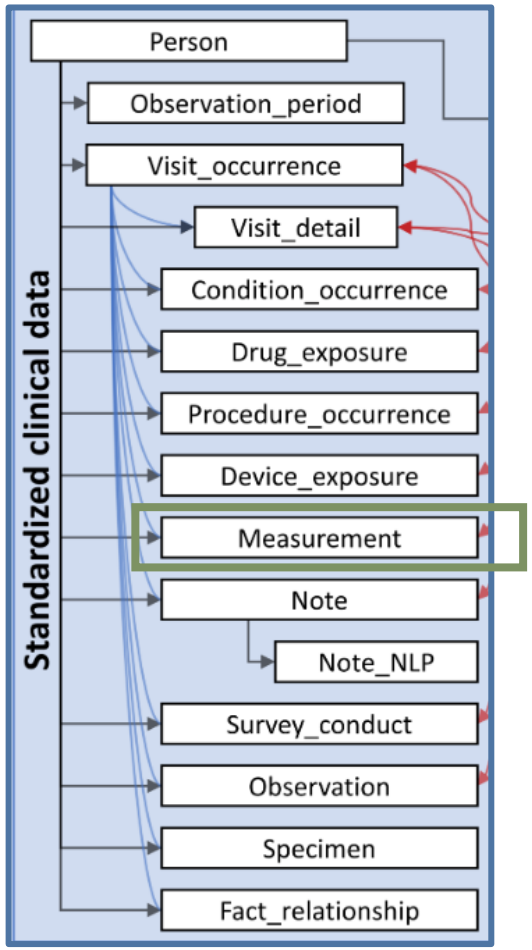
# Tablas de datos clínicos




## MEASUREMENT:

```
$ measurement_id                <int>
$ person_id                     <int>
$ measurement_concept_id        <int>
$ measurement_date              <date>
$ measurement_datetime          <dttm>
$ measurement_time              <chr>
$ measurement_type_concept_id   <int>
$ operator_concept_id           <int>
$ value_as_number               <dbl>
$ value_as_concept_id           <int>
$ unit_concept_id               <int>
$ range_low                     <dbl>
$ range_high                    <dbl>
$ provider_id                   <int>
$ visit_occurrence_id           <int>
$ visit_detail_id               <int>
$ measurement_source_value      <chr>
$ measurement_source_concept_id <int>
$ unit_source_value             <chr>
$ value_source_value            <chr>
```

Clave primaria

Clave externa a la persona

Clave externa al concepto

Clave externa a la visita

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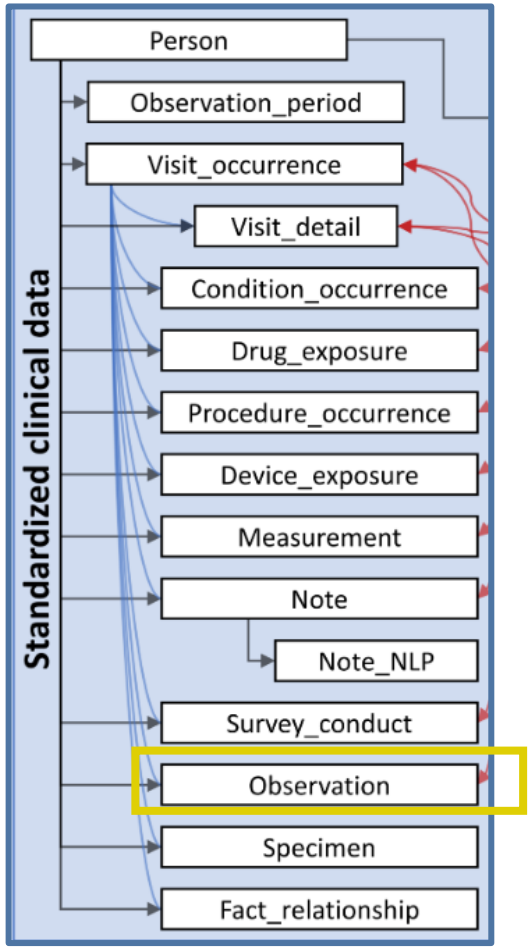
# Tablas de datos clínicos




## OBSERVATION:

```
$ observation_id              <int>
$ person_id                   <int>
$ observation_concept_id      <int>
$ observation_date            <date>
$ observation_datetime        <dttm>
$ observation_type_concept_id <int>
$ value_as_number             <dbl>
$ value_as_string             <chr>
$ value_as_concept_id         <int>
$ qualifier_concept_id        <int>
$ unit_concept_id             <int>
$ provider_id                 <int>
$ visit_occurrence_id         <int>
$ visit_detail_id             <int>
$ observation_source_value    <chr>
$ observation_source_concept_id <int>
$ unit_source_value           <chr>
$ qualifier_source_value      <chr>
```

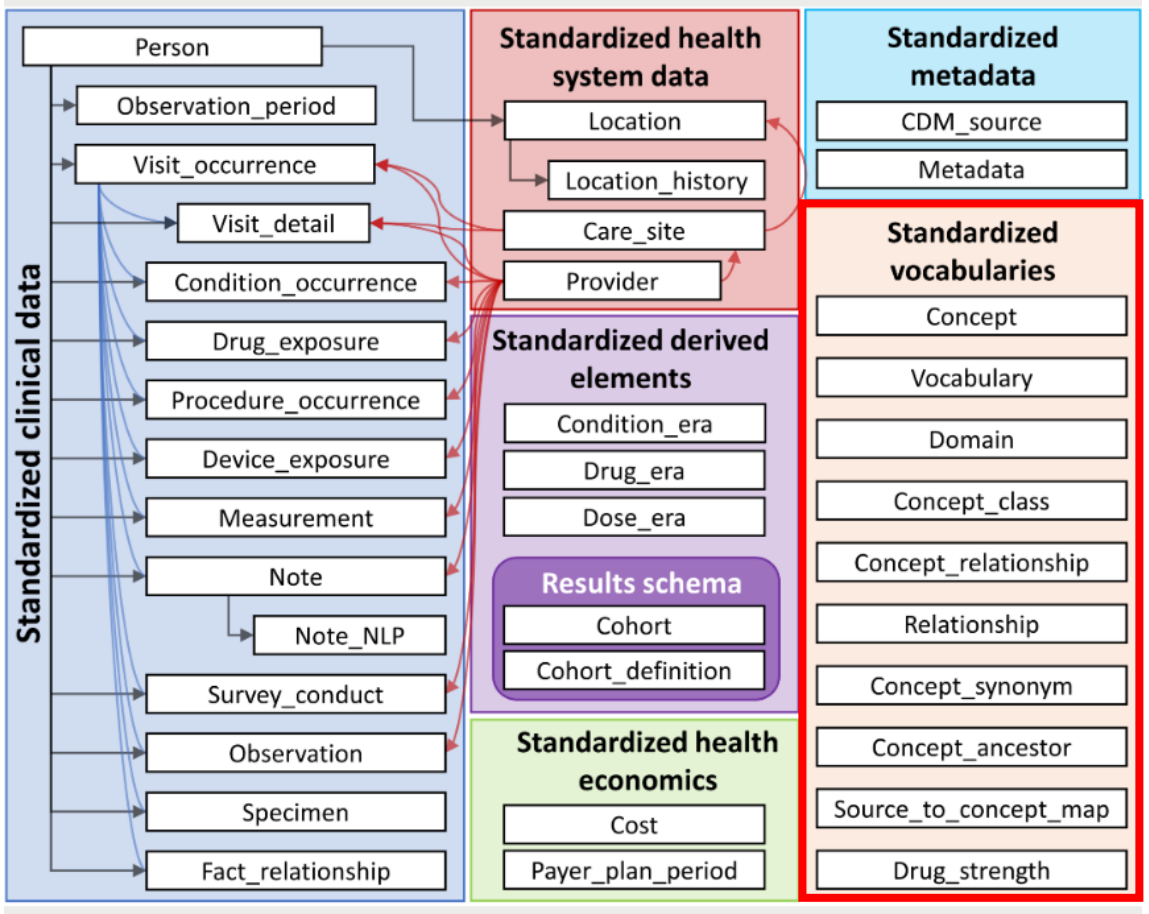
Clave primaria

Clave externa a la persona

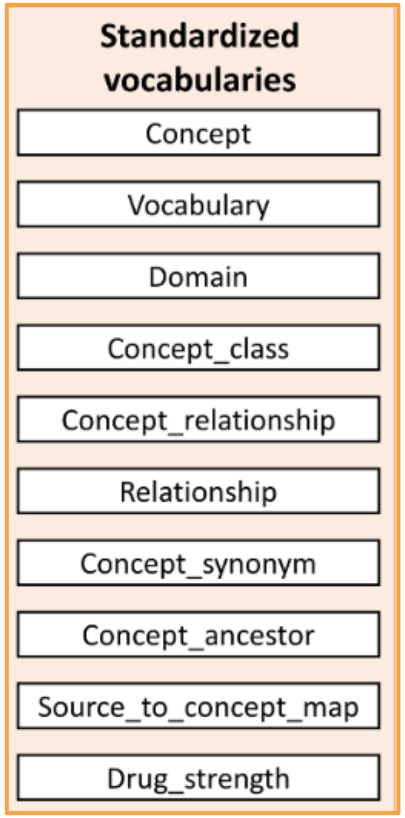
Clave externa al concepto

Clave externa a la visita

# Tablas de vocabularios en OMOP



# Vocabulario en OMOP




Los **Vocabularios Estandarizados** contienen todos los conjuntos de códigos, terminologías, vocabularios, nomenclaturas, clasificaciones, etc. que permiten:

- Generar la base de datos OMOP (estandarizada) desde la FUENTE.
- Interpretar el significado de los datos.
- Navegar entre jerarquías y relaciones entre diferentes conceptos.

**Hay un repositorio online **(ATHENA)** disponible para acceder a los vocabularios estandarizados.*

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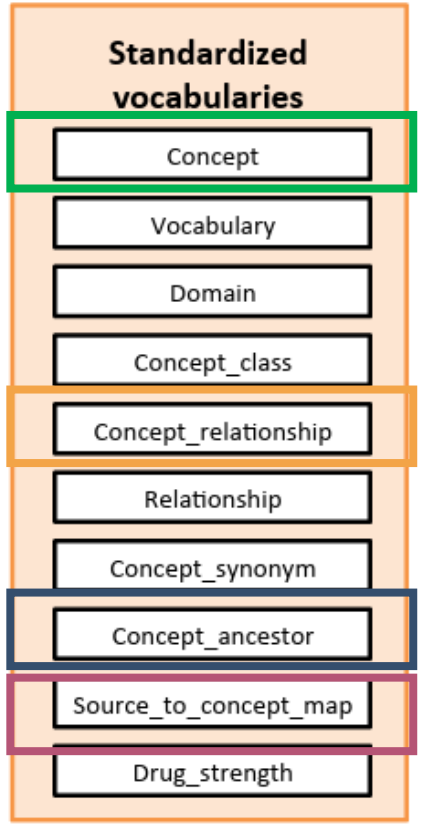
# Vocabularios en OMOP




- CONCEPTOS: describen cada una de las entidades (contenido de las tablas)
- RELACIONES: conectan conceptos (ej: debido a, mapeado desde)
- JERARQUÍA: ir desde un concepto general (padre) a más específicos (hijos).
- MAPEO: mapeo entre códigos fuentes y estándar.

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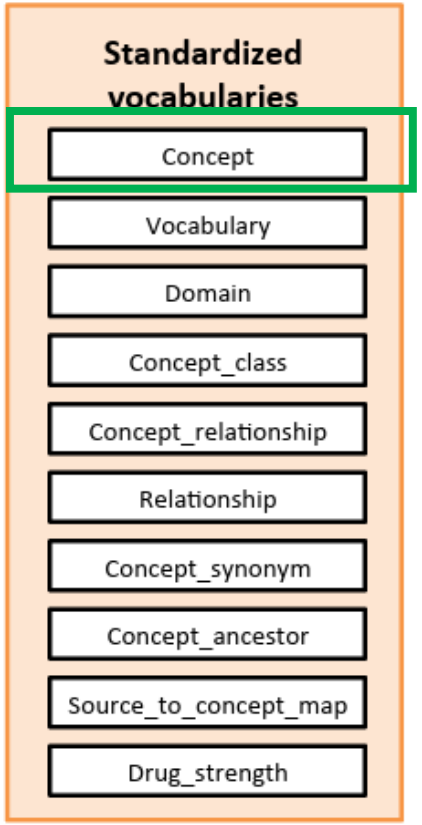
# CONCEPT




Base de la estandarización semántica en OMOP.

**CONCEPT:**

```
$ concept_id       <int> ...
$ concept_name     <chr> ...
$ domain_id        <chr> ...
$ vocabulary_id    <chr> ...
$ concept_class_id <chr> ...
$ standard_concept <chr> ...
$ concept_code     <chr> ...
$ valid_start_date <date> ...
$ valid_end_date   <date> ...
$ invalid_reason   <chr> ...
```

Clave primaria: id único del concepto

Código del concepto en su vocabulario

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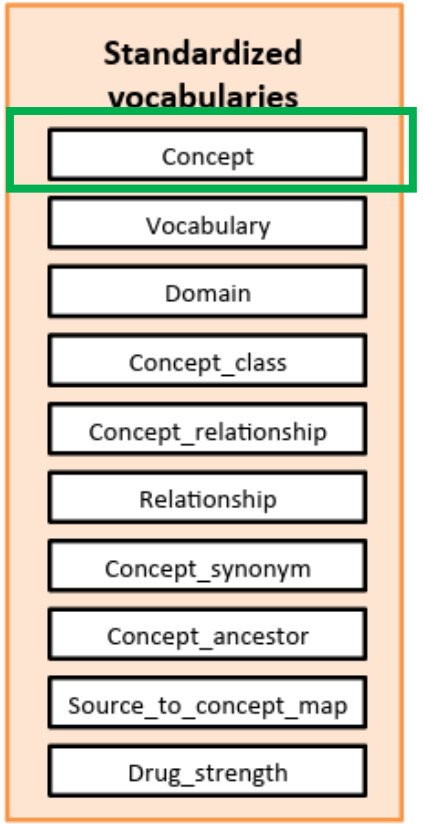
# CONCEPT




```
cdm$concept
```

| | concept_id | concept_name | domain_id | vocabulary_id | standard_concept | concept_code |
|---|---|---|---|---|---|---|
| | <int> | <chr> | <chr> | <chr> | <chr> | <chr> |
| 1 | 42618572 | Idiopathic osteoporosis … | Condition | ICD10 | NA | M80.52 |
| 2 | 45595844 | Mental and behavioural d… | Condition | ICD10 | NA | F10.1 |
| 3 | 45606094 | Hydroxyapatite depositio… | Condition | ICD10 | NA | M11.0 |
| 4 | 35626848 | Succinylcholine chloride… | Drug | SNOMED | NA | 770263009 |
| 5 | 44799055 | [X]Other corneal scars a… | Condition | SNOMED | NA | 45606100000… |
| 6 | 4070460 | Structure of lymphatic v… | Spec Ana… | SNOMED | S | 1765002 |

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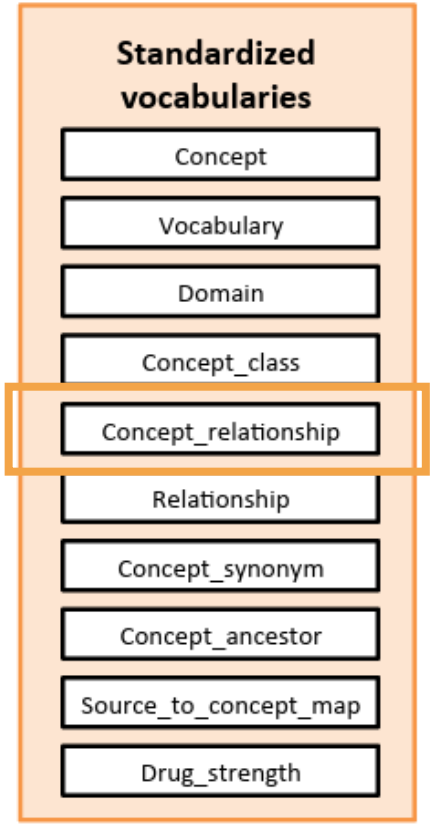
# CONCEPT_RELATIONSHIP




```
cdm$concept_relationship
```

| | concept_id_1 | concept_id_2 | relationship_id | valid_start_date | valid_end_date | invalid_reason |
|---|---|---|---|---|---|---|
| | <int> | <int> | <chr> | <date> | <date> | <chr> |
| 1 | 35804314 | 912065 | Has modality | 2021-01-26 | 2099-12-31 | NA |
| 2 | 35804314 | 42542145 | Has accepted use | 2019-08-29 | 2099-12-31 | NA |
| 3 | 35804314 | 42542145 | Is current in | 2019-08-29 | 2099-12-31 | NA |
| 4 | 35804327 | 35803584 | Has context | 2019-05-27 | 2099-12-31 | NA |
| 5 | 35804327 | 42542145 | Has accepted use | 2019-08-29 | 2099-12-31 | NA |
| 6 | 35804327 | 42542145 | Is historical in | 2019-08-29 | 2099-12-31 | NA |
| 7 | 35804329 | 35803584 | Has context | 2019-05-27 | 2099-12-31 | NA |
| 8 | 35804329 | 42542145 | Has accepted use | 2019-08-29 | 2099-12-31 | NA |
| 9 | 35804329 | 42542145 | Is historical in | 2019-08-29 | 2099-12-31 | NA |
| 10 | 35804338 | 35803584 | Has context | 2019-05-27 | 2099-12-31 | NA |

# 1 more rows

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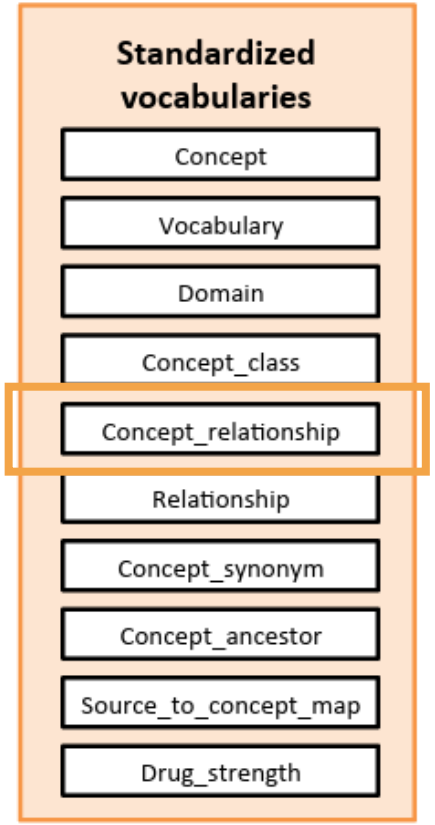
# CONCEPT_RELATIONSHIP




cdm$concept_relationship

```
cdm$concept_relationship %>%
  filter(concept_id_1 == 256451)%>%
  left_join(cdm$concept %>% select(concept_id, concept_name),
    by = c("concept_id_2"="concept_id"))
```

Bronchitis concept_id

| | concept_id_1 | concept_id_2 | relationship_id | concept_name |
|---|---|---|---|---|
| | <int> | <int> | <chr> | <chr> |
| 1 | 256451 | 4077220 | Value mapped from | H/O: bronchitis |
| 2 | 256451 | 3538386 | Concept was_a from | Bronchitis NOS |
| 3 | 256451 | 3538386 | Mapped from | Bronchitis NOS |
| 4 | 256451 | 40642537 | Has status | Defined |
| 5 | 256451 | 40642539 | Has Module | SNOMED CT core |
| 6 | 256451 | 256722 | Subsumes | Bronchopneumonia |
| 7 | 256451 | 256896 | Subsumes | Tuberculosis of bronchus |
| 8 | 256451 | 4077220 | Asso finding of | H/O: bronchitis |
| 9 | 256451 | 4148123 | Concept was_a from | Bronchitis unspecified |
| 10 | 256451 | 4148123 | Mapped from | Bronchitis unspecified |

# i more rows

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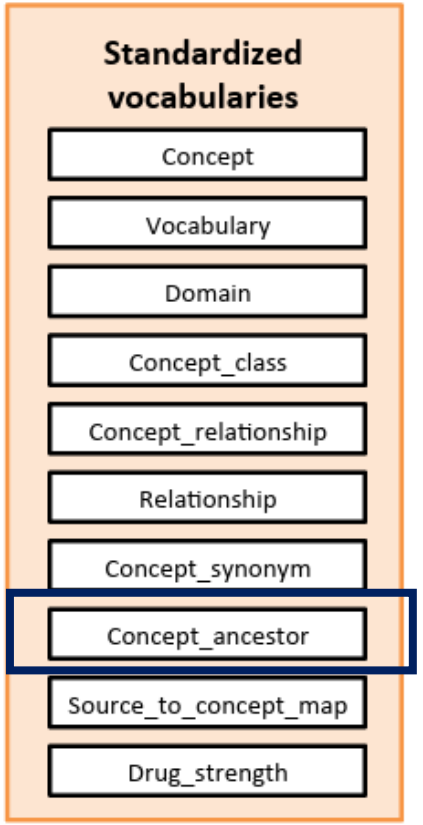
# CONCEPT_ANCESTOR




```
cdm$concept_ancestor
```

```
> cdm$concept_ancestor
# Source:   table<concept_ancestor> [?? x 4]
# Database: DuckDB 1.4.4 [agiuliodori@Windows 10 x64:R 4.4.0/C:\Users\agiuliodori\AppData\Local\
e4b1c5a746600.duckdb]
   ancestor_concept_id descendant_concept_id min_levels_of_separation max_levels_of_separation
                 <int>                 <int>                     <int>                     <int>
 1              375415               4335743                         4                         4
 2              727760               2056453                         1                         1
 3              735979              41070383                         3                         5
 4              438112              36566114                         2                         3
 5              529411              43269406                         3                         3
 6              141960                200164                         3                         6
 7              441840               4096781                         4                        12
 8              438112              42514191                         3                         3
 9              439392              36525635                         2                         2
10              438112              36533826                         5                        10
```

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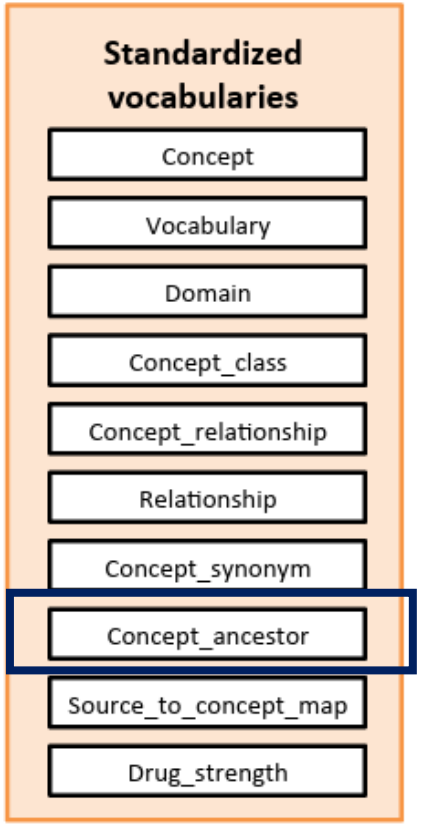
# CONCEPT_ANCESTOR




cdm$concept_ancestor

```
cdm$concept_ancestor %>%
  filter(ancestor_concept_id == 256451)%>%
  left_join(cdm$concept %>% select(concept_id, concept_name),
     by = c("descendant_concept_id"="concept_id"))
```

Bronchitis concept_id

| | ancestor_concept_id | descendant_concept_id | concept_name |
|---|---|---|---|
| | <int> | <int> | <chr> |
| 1 | 256451 | 4170141 | Acute fibrinous laryngotracheobronchitis |
| 2 | 256451 | 46269720 | Bronchopneumonia due to Pseudomonas |
| 3 | 256451 | 4197404 | Acute bronchitis with bronchospasm |
| 4 | 256451 | 46269713 | Bronchopneumonia due to Haemophilus influenzae |
| 5 | 256451 | 4245499 | Pleurobronchopneumonia |
| 6 | 256451 | 260139 | Acute bronchitis |
| 7 | 256451 | 46269714 | Bronchopneumonia due to Human metapneumovirus |
| 8 | 256451 | 46269710 | Bronchopneumonia due to Escherichia coli |
| 9 | 256451 | 4208807 | Acute infective bronchitis |
| 10 | 256451 | 37206138 | Bronchitis co-occurrent with acute wheeze |

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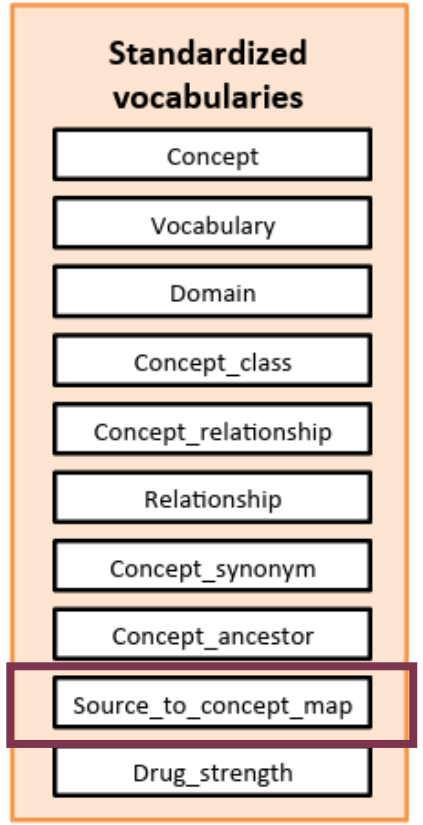
# SOURCE_TO_CONCEPT_MAP




```
cdm$source_to_concept_map
```

| | source_code | source_concept_id | source_vocabulary_id | source_code_description | target_concept_id | target_vocabulary_id |
|---|---|---|---|---|---|---|
| | <chr> | <int> | <chr> | <chr> | <int> | <chr> |
| 1 | D38.0 | 0 | ICD10 | tumor of uncertain or unknown b… | 22274 | SNOMED |
| 2 | 235.6 | 0 | ICD9CM | neoplasm of uncertain behavior … | 22274 | SNOMED |
| 3 | D57 | 0 | ICD10 | disorders sickle | 22281 | SNOMED |
| 4 | 282.60 | 0 | ICD9CM | sickle cell disease, unspecified | 22281 | SNOMED |
| 5 | D58.1 | 0 | ICD10 | el·liptocitosi hereditary | 22288 | SNOMED |
| 6 | 282.1 | 0 | ICD9CM | el·liptocitosi hereditary | 22288 | SNOMED |
| 7 | I85.9 | 0 | ICD10 | esophageal varices without blee… | 22340 | SNOMED |
| 8 | 456.1 | 0 | ICD9CM | without mention of bleeding eso… | 22340 | SNOMED |
| 9 | J38.4 | 0 | ICD10 | Laryngeal edema | 22350 | SNOMED |
| 10 | 478.6 | 0 | ICD9CM | Laryngeal edema | 22350 | SNOMED |

# ℹ more rows

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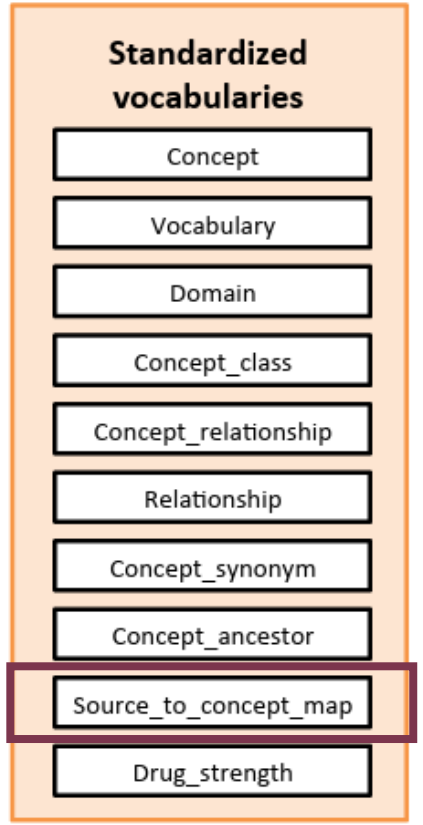
# SOURCE_TO_CONCEPT_MAP




```
cdm$source_to_concept_map
```

**CÓDIGOS FUENTE** → **CÓDIGOS ESTÁNDAR**

Ej. SIDIAP utiliza:
- **ICD10-CM**
- **ATC-7**

OMOP utiliza:
- **SNOMED**
- **RxNorm**

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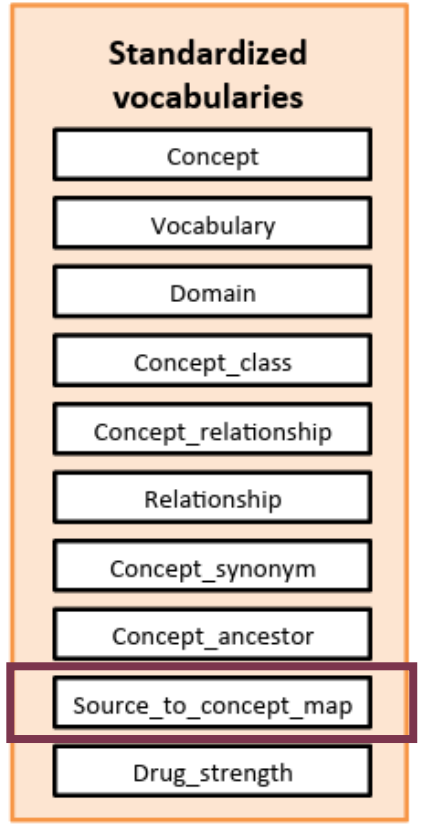
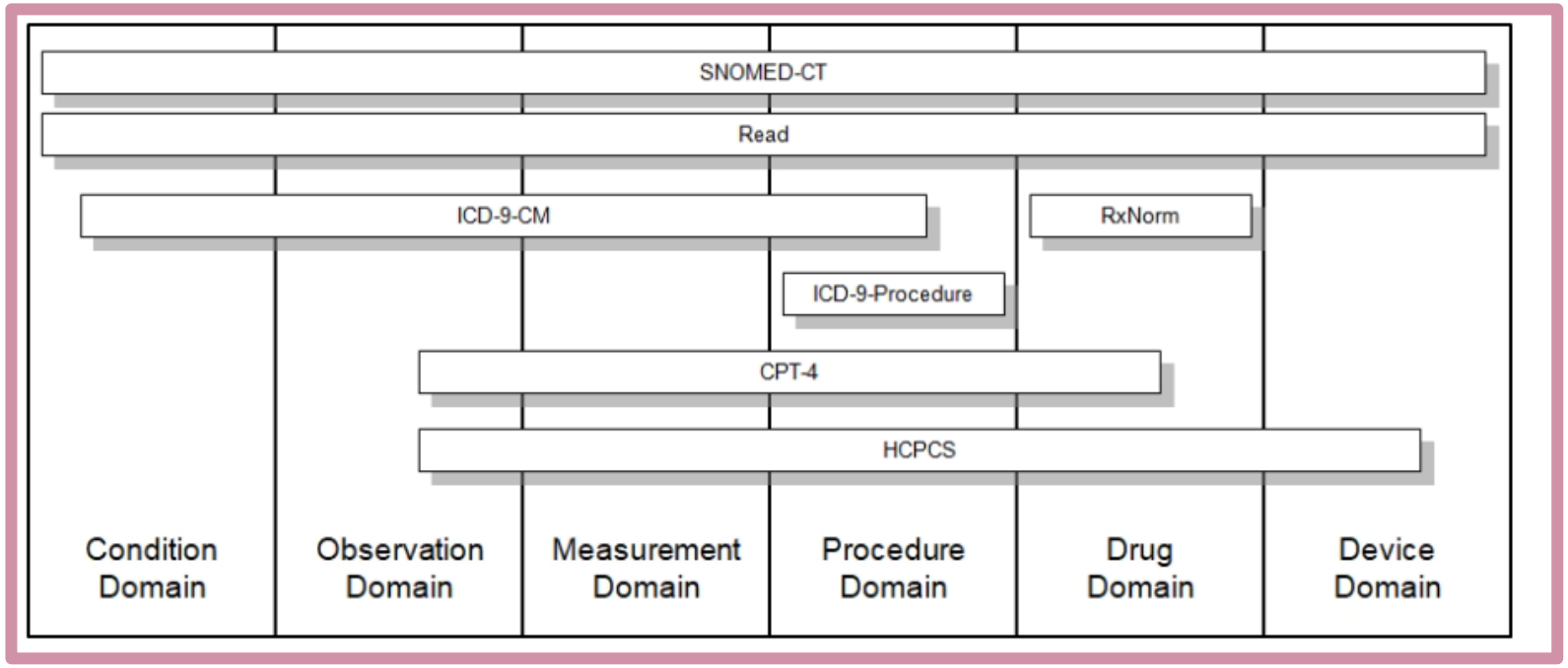
# SOURCE_TO_CONCEPT_MAP




## VOCABULARIOS ESTÁNDAR EN OMOP




https://github.com/OHDSI/Vocabulary-v5.0/wiki/General-Structure,-Download-and-Use

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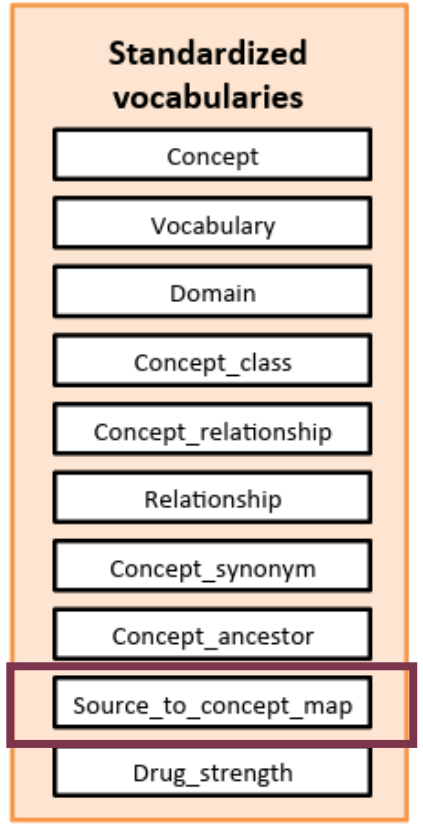
# SOURCE_TO_CONCEPT_MAP




```
cdm$source_to_concept_map
```

```
> cdm$source_to_concept_map %>% filter(source_code == "J40")
# Source:   SQL [1 x 9]
# Database: postgres
  source_code source_concept_id source_vocabulary_id source_code_description target_concept_id target_vocabulary_id valid_start_d
  <chr>                    <int> <chr>               <chr>                             <int> <chr>                <date>
1 J40                          0 ICD10               bronchitis, not classi…          256451 SNOMED               1970-01-01
# ℹ 2 more variables: valid_end_date <date>, invalid_reason <chr>
```

```
> cdm$source_to_concept_map %>% filter(target_concept_id == 256451)
# Source:   SQL [2 x 9]
# Database: postgres
  source_code source_concept_id source_vocabulary_id source_code_description      target_concept_id target_vocabulary_id
  <chr>                    <int> <chr>               <chr>                                  <int> <chr>
1 J40                          0 ICD10               bronchitis, not classified …          256451 SNOMED
2 490                          0 ICD9CM              Bronchitis not specified as…          256451 SNOMED
```

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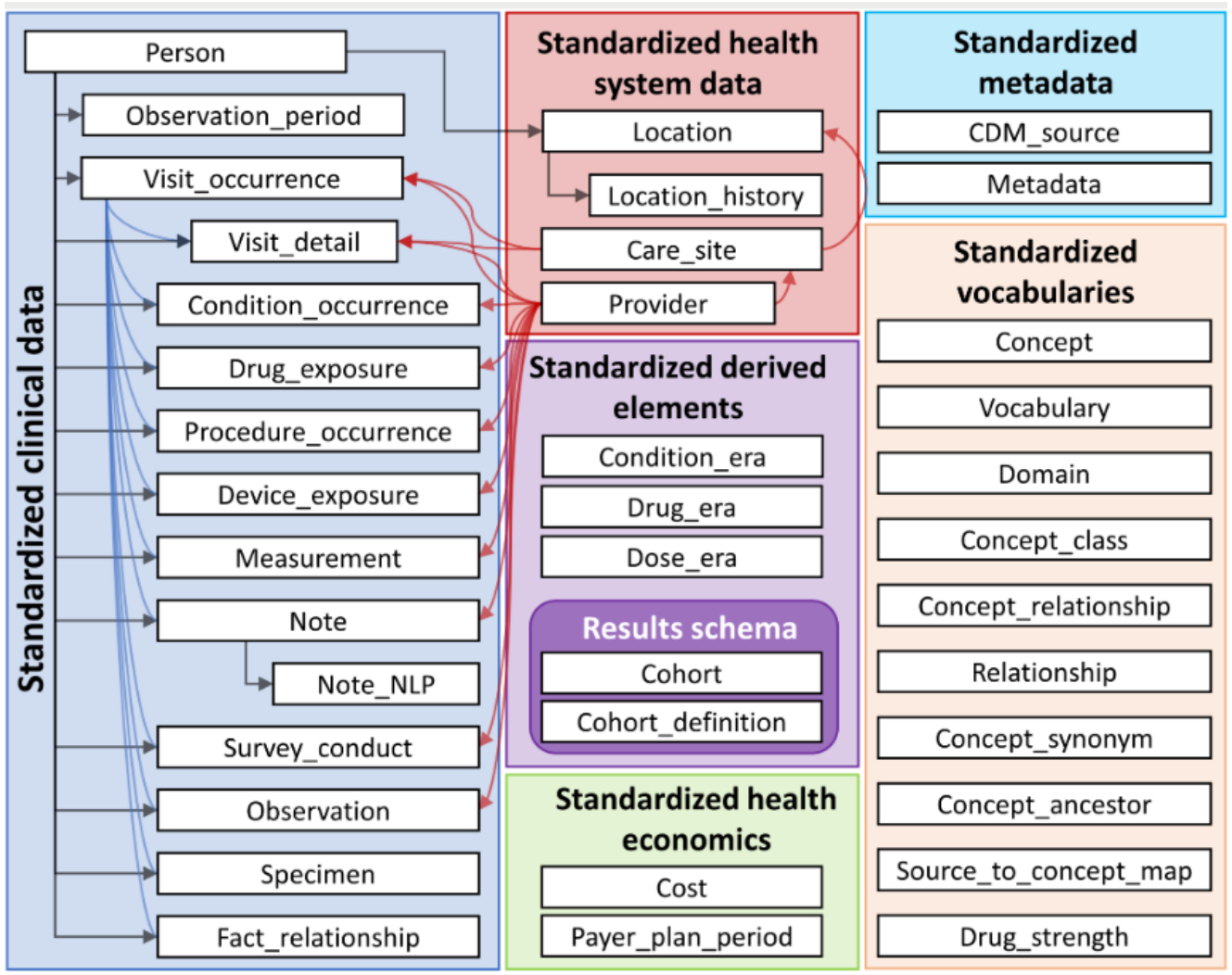
# Tablas en OMOP CDM




THE BOOK OF OHDSI

OBSERVATIONAL HEALTH DATA SCIENCES AND INFORMATICS

https://ohdsi.github.io/TheBookOfOhdsi/

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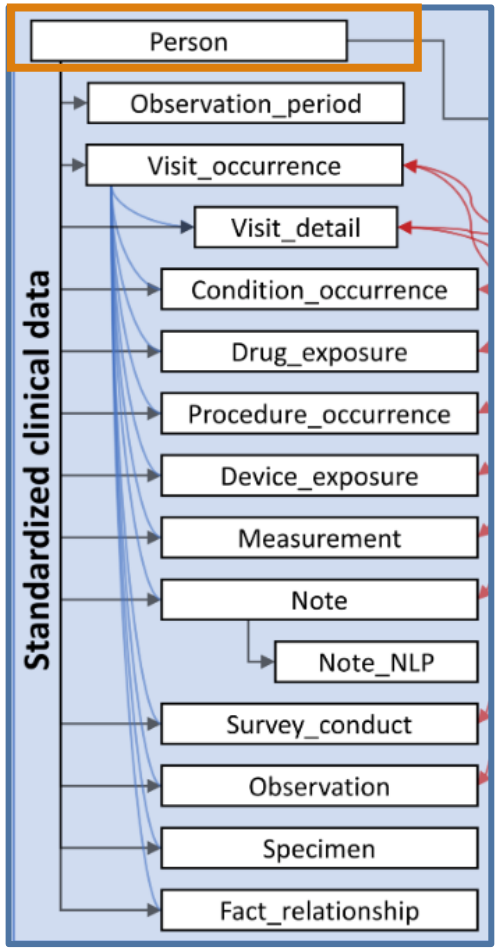
# Tablas en OMOP CDM




```
cdm$person
```

```
$ person_id                <chr> "xx", "xx", "xx", "xx", "xx", "xx", "xx", "xx", "xx"
$ gender_concept_id        <int> 8507, 8532, 8507, 8532, 8532, 8507, 8532, 8507, 8532
$ year_of_birth            <int> 1977, 1926, 1954, 1964, 1936, 1974, 1910, 1947, 1930
$ month_of_birth           <int> 3, 11, 9, 7, 7, 10, 8, 6, 6, 2
$ day_of_birth             <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
$ birth_datetime           <dttm> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA
$ race_concept_id          <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0
$ ethnicity_concept_id     <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0
$ location_id              <int> 1, 1, 1, 1, 1, 1, 9, 1, 1, 1
$ provider_id              <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA
$ care_site_id             <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA
$ person_source_value      <chr> "1", "2", "3", "4", "6", "7", "8", "9", "10", "11"
$ gender_source_value      <chr> "H", "D", "H", "D", "D", "H", "D", "H", "D", "H"
$ gender_source_concept_id <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA
$ race_source_value        <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA
$ race_source_concept_id   <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA
$ ethnicity_source_value   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA
$ ethnicity_source_concept_id <int> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA
```

# *Gracias*

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